

Analysis of the Li-Ion battery dataset

1. Business Understanding & Problem Framing

The dataset captures a battery cycling experiment — each battery goes through **charge** → **discharge** → **impedance spectroscopy** cycles. The core problem is **understanding and predicting battery degradation**, specifically:

- How does capacity fade over charge/discharge cycles?
- What does State of Health (SOH) look like over time?
- Which measurable signals (impedance, voltage, current) best predict degradation?
- Can we build a reliable SOH predictor from available features?

Target variable: **capacity** → normalized to **SOH (%)** = $\text{capacity} / \text{initial capacity} \times 100$.

2. EDA & Data Cleaning

Dataset Structure (105,128 rows × 19 cols): The dataset is interleaved across 3 experiment types — each with its own active columns:

Type	Rows	Active Columns	Purpose
charge	87,782	current_charge, voltage_charge, time	CC/CV charging protocol
discharge	16,338	current_load, voltage_load, capacity	Load behavior, capacity measurement
impedance	1,008	sense_current, battery_current, impedance, Re, Rct	EIS spectroscopy

Key EDA findings:

- **ambient_temp** = 24°C for **all 105,128 rows** — zero variance, completely useless.
- Capacity ranges from **1.768 Ah** → **1.849 Ah** across 28 discharge cycles; SOH at end = **95.7%** (battery is still healthy in this dataset).
- SOH does not reach 80% EOL threshold — the battery is in early-to-mid life.

- Cycles are **not interleaved by type** — impedance cycles (0,2,12,14...) and discharge cycles (3,5,7,9...) never overlap, requiring interpolation to build a per-cycle feature matrix.

3. Feature Engineering

Q1 — Columns to Remove?

Remove immediately:

- `ambient_temp` — single unique value (24), contributes zero information.
- `current_ratio` — **exactly** derivable as `sense_current / battery_current` (verified computationally). Redundant.

Remove contextually (type-specific nulls are structural, not missing data — keep but handle per-type):

- `sense_current`, `battery_current`, `impedance`, `rectified_impedance` are only populated for impedance rows. These are string-encoded complex numbers — usable only after parsing.
- `Re` and `Rct` are the **already-extracted real parts** of the EIS result. Since `rectified_impedance` adds little beyond `impedance` (correlation with real part is only -0.52), and `Re/Rct` are cleaner scalar representations, `rectified_impedance` can be dropped for modeling.

Q2 — Expanding with New Features?

Yes — several high-value features were engineered:

New Feature	Derived From	What It Captures
<code>SOH</code> (target)	<code>capacity / initial_capacity</code>	Normalized battery health
<code>total_resistance</code>	<code>Re + Rct</code>	Overall internal resistance — single aging proxy
<code>Re_Rct_ratio</code>	<code>Re / Rct</code>	Balance between electrolyte and charge-transfer degradation
<code>imp_magnitude</code>	<code>abs(impedance)</code>	impedance

<code>imp_phase</code>	$\angle \text{impedance}$ (degrees)	Phase angle — sensitive to SEI layer growth
<code>discharge_duration</code>	$\max(\text{time}) - \min(\text{time})$ per discharge cycle	How long the battery can sustain discharge load
<code>voltage_drop</code>	$V_{\text{first}} - V_{\text{last}}$ per discharge	Voltage sag under load
<code>discharge_energy</code>	$\Sigma(I \times V)$ per discharge	Total energy delivered
<code>Re_lag1,</code> <code>Rct_lag1</code>	Shifted 1 cycle	Previous cycle state — temporal context
<code>Rct_delta,</code> <code>Re_delta</code>	Current - previous	Rate of change of degradation

Q3 — Combining Columns to Create "Better" Columns?

A "better" column is one that is **more directly predictive, physically meaningful, and lower-noise** than its components separately. Three combinations stand out:

`total_resistance = Re + Rct` — Both `Re` (electrolyte resistance) and `Rct` (charge-transfer resistance) increase with aging, but for different reasons. Their sum gives a single scalar for internal resistance growth, which is strongly anti-correlated with capacity.

`imp_magnitude = |Re_part + j·Im_part|` — The raw `impedance` column stores complex numbers as strings (e.g., "`0.143-0.119j`"). Parsing and taking magnitude collapses both components into one physically interpretable metric (total AC impedance), cleaner for ML.

`Rct_delta` (cycle-over-cycle change in `Rct`) — `Rct` in isolation tells you the current state; `Rct_delta` tells you the **rate of degradation**. This ranked in the top 5 features for Random Forest and captures the trajectory of aging, not just the snapshot.

4. Modelling & Evaluation

With only 28 discharge cycles (27 after lag features), this is a **small-N problem**. A per-cycle feature matrix was built by interpolating impedance measurements to discharge cycle numbers.

Model	MAE (SOH %)	R ²
Linear Regression	2.93%	-93.9
Ridge Regression	1.51%	-25.3
Gradient Boosting	0.64%	-0.62
Random Forest	0.86%	-1.51

Interpretation: R² values are negative — this is expected with only 6 test points and very low SOH variance (95.7%–100%). The model is essentially predicting near-constant values in a narrow band. MAE of 0.64% for Gradient Boosting means predictions are within ~0.6 percentage points of actual SOH, which is practically useful. The negative R² signals **underfitting due to dataset size**, not model failure.

Top predictive features (Random Forest):

1. `discharge_duration` (0.75) — most powerful single signal
2. `max_voltage_charge`, `mean_voltage_load` — voltage profile shape
3. `imp_phase`, `Rct_delta` — EIS-derived aging signals

5. Insights & Recommendations

Key Insights:

- **Discharge duration is the single strongest predictor of SOH** — as the battery ages, it can sustain discharge for shorter periods even under the same load. This is a non-invasive, easily observable signal.
- **Rct grows faster than Re** — charge-transfer resistance is the more sensitive aging indicator. Rct increased ~63% over the experiment while Re changed <10%. Track Rct specifically.
- **The battery is still healthy** — SOH never dropped below 95.7%, meaning this dataset represents early-life degradation. Models trained here will not generalize to end-of-life behavior without more cycles.
- **The `impedance` column contains rich complex-valued EIS data that is wasted in string form.** Parsing it into magnitude and phase doubles the information extracted from the EIS experiment.

Recommendations:

1. **Collect more cycles** (80+) to observe the accelerating degradation phase and train robust SOH prediction models.
2. **Use `total_resistance` and `Rct_delta` as real-time health KPIs** — both are low-cost to compute and physically grounded.
3. **Standardize `ambient_temp`** — if future experiments vary temperature, this becomes a crucial covariate. Currently it is noise.
4. **Replace the string-encoded complex columns** with their parsed real, imaginary, magnitude, and phase at the data collection stage. The current format causes significant preprocessing overhead.
5. **For deployment**, Gradient Boosting with lag features is recommended — it handles the temporal structure and nonlinear relationships between impedance components and capacity best.